Guillain–Barré Syndrome

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Definition

Guillain–Barré syndrome (GBS) is an acute onset inflammatory polyneuropathy characterized by rapidly progressive, symmetric, ascending weakness with areflexia in a previously normal child. GBS is the most common cause of acute flaccid paralysis in childhood.

Incidence

The annual incidence of GBS is 0.3–1.3 cases per 100,000 persons <18 years of age.

Clinical Signs and Symptoms

- ☑ Guillain–Barré syndrome presents with acute onset weakness, difficulty in walking and rising from sitting position, fatigue, pain, and paresthesia.
- ☑ Weakness is often symmetrical; starting distally in the lower limbs and progressively involves upper limbs, respiratory muscles, and cranial nerves.
- ☑ About 40–50% patients with GBS may have facial weakness and nearly about 30% have respiratory weakness.
- ☑ Weakness in GBS is rapidly progressive, reaches nadir within 2 weeks in 80% cases and in all by 4 weeks.
- ☑ Pain in the legs, thighs, buttocks, back, and neck with neck rigidity and meningismus can occur due to nerve root inflammation.
- ☑ Hypertension, blood pressure fluctuations, sinus tachycardia, and abnormal sweating suggest autonomic dysfunction and may be observed in up to 50% patients with GBS.

There are several GBS subtypes based on the evolution of clinical features and nerve conduction findings **(Table 1)**. Most common subtypes are acute inflammatory demyelinating polyneuropathy (AIDP) and acute motor axonal polyneuropathy (AMAN) variants. AMAN variant is common in Southeast Asia and AIDP in North America and Europe.

TABLE 1: Clinical variants of GBS with key clinical features and NCS findings.

Type	***Clinical features***	***Associated antibodies***	***NCS findings***
Acute demyelinating inflammatory polyneuro-pathy (AIDP)	Symmetric ascending paralysis and proximal weakness	GM1, GM4, GD1a, and GD1b	Slow CV and prolonged DL
Acute motor axonal neuropathy (AMAN)	Symmetric ascending paralysis and distal weakness	GM1, GM2, GD1a, and GD1b	Reduced amplitude, normal CV, and DL
Acute motor and sensory axonal neuropathy (AMSAN)	Axonal motor and sensory	GD1a	Axonal motor and sensory polyneuropathy
Miller Fisher syndrome (MFS)	Ophthalmoparesis, ataxia, and areflexia	GQ1b and GT1a	Axonal polyneuropathy
Bickerstaff's brainstem encephalitis	Ophthalmoparesis, ataxia, and areflexia with long tract signs	GQ1b and GT1a	Axonal polyneuropathy
Pharyngeal-cervical-brachial variant	Descending with neck and shoulder weakness	GT1a, GQ1b, and GD1a	Axonal polyneuropathy
Polyneuritis cranialis	Bifacial weakness with dysphagia and dysphonia	GQ1b and GT1a	Demyelinating polyneuropathy
Acute (ataxic) sensory neuropathy	Acute onset ataxia	GQ1b and GT1a	Axonal polyneuropathy
Acute pandysautonomia	Acute hypertension, posterior reversible encephalopathy syndrome (PRES), sweating abnormalities, and tachycardia		Axonal polyneuropathy
Acute ophthalmoparesis	Isolated ophthalmoparesis	GQ1b and GT1a	Axonal polyneuropathy

(CV: conduction velocity; DL: distal latency; GBS: Guillain–Barré syndrome; NCS: nerve conduction studies)

Guillain–Barré syndrome is an immune-mediated disorder resulting from activation of T cells and production of antibodies directed at the protein antigens located on the peripheral nerve sheath proteins. Infection with Epstein–Barr virus, *Cytomegalovirus*, *Mycoplasma pneumoniae*, and *Campylobacter jejuni*; recent immunization; and surgery may trigger formation of antibodies. About 10–20% of children with GBS can have antiganglioside antibodies **(Table 1)**.

Antibodies in patients with AIDP are directed against protein antigens on the myelin sheath while antibodies are directed against the target antigens at or near nodes of Ranvier in patients with AMAN variant **(Fig. 1)**. Antiganglioside antibodies (anti-GM1, anti-GD1a, etc.) bind to the nodal axolemma, forms membrane attack complexes (MACs), leading to complement pathway activation and disappearance of voltage-gated sodium channels. Damage at the nodes can lead to detachment of paranodal myelin sheath and nerve conduction failure.

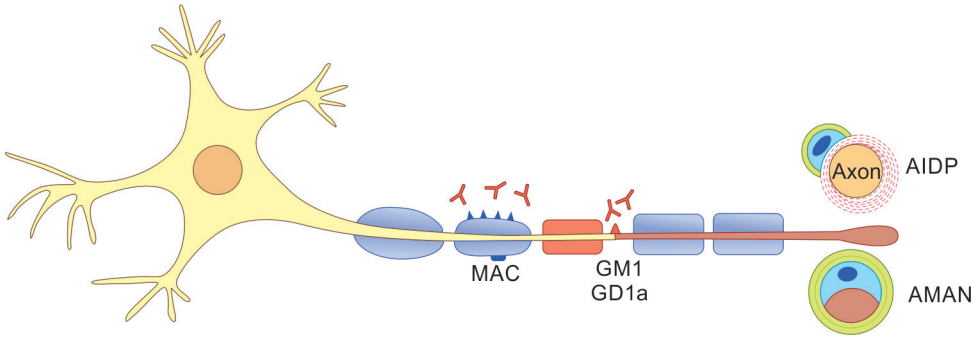

Fig. 1: Schematic diagram representing the immune-mediated pathogenesis of GBS subtypes. (AIDP: acute inflammatory demyelinating polyneuropathy; AMAN: acute motor axonal polyneuropathy; GBS: Guillain–Barré syndrome; MAC: membrane attack complex)

- ☑ No single investigation can support or refute the diagnosis of GBS early in the course.
- ☑ The diagnosis of GBS is supported by lumbar puncture, electrophysiological studies, imaging, and/or ancillary investigations, with exclusion of alternative diagnoses **(Table 2)**.
- ☑ Presence of internal ophthalmoplegia, sensory level, persistent asymmetry, bladder involvement, cerebrospinal fluid (CSF) cell count > 50 mononuclear cells, and predominant polymorphonuclear cells are atypical for GBS.

TABLE 2: Laboratory investigations supporting the diagnosis of GBS.

Laboratory investigations and their utility	***Typical findings***
Cerebrospinal fluid analysis: ☑ Required to rule out infectious conditions ☑ Presence of albumin-cytological dissociation (ACD) support GBS diagnosis ☑ ACD can be absent in first week of illness	☑ *ACD:* Presence of <5 mononuclear cells/µL and protein >45 mg/dL within 3 weeks of symptom onset ☑ Cells >50 mononuclear cells atypical for diagnosis of GBS*
Nerve conduction studies (NCS): ☑ NCS confirms the diagnosis ☑ Identified GBS subtypes ☑ Virtually positive in 90% cases by second week ☑ Must have these finding in two or more motor nerves	*AIDP*: ☑ Slowing of motor and sensory nerve conduction velocity (<70% of lower limit of normal values for age) ☑ Prolonged distal latencies (>150%) ☑ Conduction block [proximal to distal compound muscle action potential (CMAP) ration <0.7] ☑ Temporal dispersion of CMAP ☑ Increased F and H response latencies (>120% of upper limit of normal) *AMAN and AMSAN:* ☑ Lack of neurophysiologic evidence of demyelination ☑ With reduced CMAP amplitude of motor and/or sensory nerve action potential (<80% of the lower limit of normal values for the age)
MRI spine: ☑ Not routinely indicated ☑ Required to rule of infectious or transverse myelitis or compressive myelopathy in presence of bladder dysfunction and sensory level	Shows postcontrast enhancement of nerve roots and cauda equina in GBS
Antiganglioside antibodies	GM1, GM2, GM3, GD1a, GD1b, GT1b, and GQ1b IgG and IgM can be positive in 30% cases with GBS

*>50 cells mononuclear cells may be seen in HIV, polio, enterovirus, rabies infection, and tubercular arachnoiditis

(AIDP: acute inflammatory demyelinating polyneuropathy; AMAN: acute motor axonal polyneuropathy; AMSAN: acute motor and sensory axonal neuropathy; GBS: Guillain–Barré syndrome; HIV: human immunodeficiency virus; IgG: immunoglobulin G; IgM: immunoglobulin M)

- ☑ Immunotherapy should be started if patients have rapid progression of weakness, inability to walk independently for 10 meters (Hughes disability score ≥3), presence of swallowing dysfunction, respiratory insufficiency, and severe dysautonomia.
- ☑ Intravenous immunoglobulin or plasma exchange must be initiated within 2 weeks and 4 weeks, respectively after the onset of weakness in patients with GBS.
- ☑ Treatment for GBS should be considered early among rapidly progressive cases and should not be delayed pending confirmation with nerve conduction studies (NCS).
- ☑ Intravenous immunoglobulin (IVIg) should be administered at a dose of 400 mg/kg/day for 5 days (total dose 2 g/kg) and plasma exchange (200–250 mL plasma/kg body weight, 5 alternate day sessions) are equally efficacious in management of patients with GBS. Plasma exchange followed by IVIg is not more effective than either IVIg or plasma exchange alone.
- ☑ IVIg is easier to administer and widely available than plasma exchange.
- ☑ The adverse effects of IVIg and plasma exchange are comparable.
- ☑ Corticosteroids showed no significant benefit in patients with GBS and treatment with oral corticosteroids has shown to have a negative effect on outcome.

Insufficient Response or Nonresponders

Disease progression does not imply that the treatment is ineffective, as progression might have been worse without therapy. Clinicians may consider repeating the treatment or changing to an alternative treatment, but at present no evidence exists that this approach will improve the outcome.

Treatment-related Fluctuation

- ☑ Treatment-related fluctuation (TRF) may occur in <10% of patients with GBS.
- ☑ TRF is defined as progression of symptoms within 2 months following an initial improvement or stabilization after treatment.
- ☑ Patients with TRF might benefit from repeating a full course of IVIg or plasma exchange.

Treatment

Additional Management in Patients with Guillain–Barré Syndrome in Acute Phase and Long-term

Acute complications: Pain, delirium, respiratory weakness, aspiration pneumonia, corneal ulceration, inadequate nutrition, constipation, urinary retention, infection, pressure sores, deep vein thrombosis, and limb contractures.

Long-term complications: Residual motor and sensory deficits, pain, fatigue, and psychosocial stress.

Management of GBS is best addressed with the presence of a multidisciplinary team that comprises of clinicians, nurses, physiotherapists, occupational therapists, speech therapists, physiatrists, and dietitians. Children with GBS must be assessed in acute phase at regular intervals to look for the progression of weakness and early identification of complications. Indications for pediatric intensive care unit (PICU) admission and bedside assessment for children with GBS are summarized in **Table 3**.

TABLE 3: Bedside assessment and indication for pediatric intensive care unit (PICU) admission.

Bedside examination in Guillain–Barré syndrome (GBS) patients	***Indications for PICU admission***
☑ Monitoring of single breath count ☑ Identify use of accessory respiratory muscles ☑ Assessment of muscle power using Medical Research Council (MRC) grade ☑ Assess swallowing dysfunction, gag reflex, and cough reflex ☑ Assessment of functional disability using GBS disability scale ☑ Monitor heart rate, rhythm, and blood pressure ☑ Monitor bladder and bowel function	☑ Rapidly progressive weakness ☑ Respiratory insufficiency ☑ Bulbar dysfunction ☑ Severe autonomic dysfunction

Mechanical Ventilation in Guillain–Barré Syndrome

- ☑ The need for mechanical ventilation in children with GBS aged >6 years may be predicted using Erasmus GBS Respiratory Insufficiency Score (EGRIS) **(Table 4)**.
- ☑ This score is based on three measures namely time lapse between onset of weakness and hospitalization, facial and/or bulbar weakness, and Medical Research Council (MRC) sum score at hospitalization.

TABLE 4: Assessment of Erasmus GBS Respiratory Insufficiency Score (EGRIS).

Measure of assessment	***Scoring***
Time lapse between onset of weakness and hospital admission	>7 days = 0 3–7 days = 1 <3 days = 2
Facial and/or bulbar weakness at admission	Absent = 0 Present = 1
Medical Research Council (MRC) sum score at admission*	60–51 = 0 50–41 = 1 40–31 =2 30–21 = 3 ≤20 = 4

*The MRC sum score is calculated by using the sum of muscle power scoring of the following muscles on both sides: shoulder abduction, elbow flexion, wrist extension, hip flexion, knee extension, and ankle dorsiflexion.

The EGRIS score of 0–2 indicates low risk, 3–4 indicates moderate risk, and ≥5 indicates high risk for mechanical ventilation. Autonomic dysfunction has also been found to be predictor of mechanical ventilation in childhood Guillain–Barré syndrome (GBS).

Management of Comorbidities

- ☑ Fatigue has been described in 60–80% of patient with GBS and may be managed with a graded and supervised exercise program.
- ☑ Pain in acute phase is typically radicular pain and may be explained by the nerve root inflammation.
- ☑ Chronic pain has been described in 10% of patients with GBS. It may be explained by the radicular pain, muscle pain, painful paresthesia, and arthralgia resulting from immobility or persistent damage or regeneration of nerve fibers. Pain in patients with GBS may be managed by use of nonsteroidal anti-inflammatory drugs, gabapentin, or carbamazepine.

Prognosis

- ☑ Though many children with GBS have good clinical recovery, appropriate rehabilitation program should be initiated to decrease the disability in acute phase and to reduce the long-term motor and sensory deficits and fatigue.
- ☑ The functional motor deficits at discharge may be assessed using Hughes scale **(Table 5)**.
- ☑ The respiratory complications and autonomic dysfunction may pose of risk of mortality in children with GBS, and the reported mortality in patients with GBS varied from 3–10%.
- ☑ The risk of recurrence in GBS is rare, and the risk varies from 2 to 5%.

TABLE 5: Guillain–Barré syndrome (GBS) disability score (Hughes disability score).

0	A healthy state
1	Minor symptoms and capable of running
2	Able to walk 10 meter or more without assistance but unable to run
3	Able to walk 10 meter across an open space with help
4	Bedridden or chairbound
5	Requiring assisted ventilation for at least part of the day
6	Dead

Further Reading

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